Reviewer report for manuscript ”The Conditioning Bias in Binary Decision Trees and Random Forests and Its Elimination“

General remarks

The paper deals with the fact that different implementations of decision trees and random forests differ in the way new data in prediction are assigned to the child nodes. Some implementations assign an observation to the left child node if the value of the observation is less than or equal to the split point, and others assign an observation to the left child node if the value of the observation is less than the split point, where the split points are always midpoints between neighboring values in the training data. This discrepancy in the way predictions are made can lead to different predictions between different implementations if there are variables in the data that have equidistant values. Although the differences observed are very small, the issue is still of some relevance given the popularity of trees and forests.
However, the authors' solution to the problem is questionable and the analysis performed is incorrect and biased. Please see my detailed comments below.

Major revisions

1) The authors argue that selecting the conditioning operator using cross-validation-based tuning is not viable because the tree fitted to the entire dataset behaves differently from the trees fitted within cross-validation.
   They further argue that this would be the reason why selecting the conditioning operator using cross-validation would lead to overfitting.
   However, this argument is not clear to me. First, the fact that prediction models fitted within cross-validation behave differently than those fitted to the whole dataset is not limited to the present situation. Instead, this can happen for any reasonably complex prediction method, due to known or unknown properties of the prediction method. And one does not evaluate all such properties before deciding to estimate the predictive performance of a prediction method using cross-validation.
   Similarly, it is not at all clear to me why selecting the conditioning operator using cross-validation could lead to a relevant degree of overfitting. If it is indeed the case that none of the conditioning parameters is preferable, as the authors argue ("there is no reason to assume that there is anything other than a 50% chance that the default operator is not detrimental to performance"), then cross-validation-based tuning would randomly select one of the conditioning parameters, namely the one that is minimally better fit to the data.
   But why would such a small feature of the prediction method lead to a relevant degree of overfitting?
   And if it is indeed the case that neither of the conditioning parameters is preferable, as the authors argue, then the strategy of trying both and averaging the prediction would not make sense, because then it would not matter which of the two conditioning

operators one chooses - and it would be fine to just go with the conditioning operator that the respective implementation uses.
Reading further it seems that I misunderstood the authors and by "there is no reason to assume that there is anything other than a 50% chance that the default operator is not detrimental to the performance" they mean that *a priori* it is not known which of the two conditioning operators is preferable, but that for a given dataset one of the two will lead to better performance. In this case, however, it would be advantageous to perform cross-validation-based tuning, as explained below. While it may be true that due to the fact that the thresholds within cross-validation behave slightly differently than those on the full datasets, it is still expected that the behavior of the trees fitted within cross-validation will be similar to the behavior of the tree fitted on the full dataset. Thus, one can also expect that the more suitable conditioning operator will be associated with a lower cross-validation error than the less suitable conditioning operator. In the worst case, the behavior of the trees fit within cross-validation is so different from the behavior of the tree fit on the whole data set that the tuning provides no information about the more appropriate conditioning operator. In this case, the tuning would randomly select one of the two conditioning operators. However, this would not lead to a worse prediction model than averaging the predictions over the two conditioning operators, as the authors suggest. There is no (apparent) reason why cross-validation-based tuning would select the less suitable conditioning operator if the two conditioning operators are associated with noticeably different performance. If the authors are aware of such a reason, they need to explain it precisely.
In summary, there is no reason why cross-validation-based selection of the conditioning parameter should not be considered. Therefore, the authors should include this option in their comparative studies.

2) Page 12: "What we cannot do is assume that if one conditioning performs better on a random 80% of the data, it remains superior when using all available data for training."

   One can just use a large K in K-fold cross-validation to avoid this issue. For example, when using K=10 or K=20 instead of K=5, one can assume that the performance within cross-validation is very similar than when using the whole dataset for training because it does not really matter if a small fraction of the data is removed before training (as is the case with cross-validation with a large K).

3) Page 8: "Specifically, carrying out inference with both operators and averaging their predictions effectively eliminates the dependence on the conditioning operator. This method can be expected to yield predictions at least as good as those produced by the conditioning operator with the worse performance."

   But this method will also yield predictions that are worse than when using the better suitable conditioning operator. The expected prediction performance is the same when using this averaging performance as when just randomly picking one of the conditioning

operators, that is, the proceeding implicitly done in practice when just going with the conditioning operator offered by the respective implementation.
It is much better to try to identify the better suitable conditioning operator and this can be done using cross-validation based tuning.

4) Page 12: "The objective of model selection is to identify hyperparameters that maximize the performance with the default conditioning operator (≤) of the sklearn (Pedregosa et al., 2011) implementation. Model selection was carried out individually for each dataset using 20 times repeated 5-fold cross-validation and maximizing the ROC AUC score (Fawcett, 2006) for classification and the r2 score for regression. The hyperparameters considered were the minimum number of samples on the leaf nodes (ranging from 0.5% to 20% of the dataset size) and the maximum depth (ranging from 2 to 15). The results of the model selection, presenting the optimal hyperparameters and the corresponding AUC and r2 scores, are detailed in Table 1. All subsequent results reported in the paper are achieved using these selected hyperparameters."

   There are two problems here:

   1. Selecting the tuning parameter using cross-validation and then performing cross-validation again using the selected tuning parameter leads to overoptimistic performance estimates because in this way the selected tuning parameter values depend on the test sets in the second cross-validation. The correct way for estimating the performance would be to perform a nested cross-validation where hyperparameter selection is performed in the inner loop and performance estimation is performed in the outer loop. The authors should either perform a nested cross-validation for estimating the performance or just do not perform hyperparameter selection, fixing these values to the default values. The latter would be more appropriate because in practice the tuning parameters of trees and forests are often not tuned because the performance is not very sensitive to their values.

   2. Given that, as explain in 1., the performance estimates of the default conditioning operator are overoptimistic (in the currently used procedure), the comparison to the performance estimates of the non-default conditioning operator are biased. This could explain why in Table 2 most r_d^2 values are negative, which is surprising given that one would expect that the default conditioning operator performs equally frequent better as worse compared to the non-default conditioning operator. Of course, this also leads to an optimistic bias in the already optimistically biased significance tests (see comment further down). The same applies to Table 3, where the proposed approach is a mixture of the two conditioning operators. This mixture is also optimistically biased in the comparison to the worse conditioning operator due to the optimistic bias of the predictive performance of the better conditioning operator.

5) For each dataset, the authors perform Wilcoxon signed-rank tests to test for pairwise differences in the performance estimates of different methods. Here, they perform 400

times repeated 5-fold cross-validation and treat the resulting 2000 performance scores (from the cross-validation iterations) as independent observations in the testing. However, this proceeding is highly incorrect. The 2000 performance scores within the cross-validation repetitions are not independent as the training and test sets strongly overlap between the different cross-validation iterations. However, it is a crucial assumption of the Wilcoxon signed-rank test that the pairs of observations (i.e. performance scores) are independent from each other. By incorrectly assuming that the performance scores are independent, it is not surprising that many of the differences are seemingly statistically significantly different because for the tests the apparent (not real!) sample sizes are very large (n=2,000).
The fact that the authors' proceeding is highly incorrect can also be seen by realizing that had they further increased the number of cross-validation repetitions, the apparent sample size would have been even larger. For example, if they had performed 10,000 repetitions of the cross-validation the apparent sample size would have been n=50,000, making it likely that even the smallest differences would have been -incorrectly- statistically significant.
Given that there is no unbiased estimator for the variance of K-fold cross-validation (Bengio & Grandvalet, 2003), it is not possible to test for differences between cross-validation-based performance estimates for single datasets. Therefore, the authors must remove the significance testing in order to prevent reporting incorrect and misleading results.

Reference:

Bengio, Y., & Grandvalet, Y. (2003). No Unbiased Estimator of the Variance of K-Fold Cross-Validation.

6) Tables 2 and 3 should also show the absolute AUC and r_d^2 values for both conditioning operators and not only the difference.

7) Given that, as described, the analysis is strongly flawed, it is not possible to tell whether the conclusions drawn from it are valid.

8) It also does not make sense to specifically highlight the greatest performance differences achieved across datasets because it is clear that the maximum differences depend on the number of datasets included, which is why they do not provide well-interpretable information.

Minor essential revisions

1) The introduction discusses trees in a very general form. While this is interesting, it is beyond the scope of the paper, which I why the introduction should be shortened considerably

2) In the abstract the approach proposed by the authors is not yet described, even though it is very simple. It should be described already there for clarity.

3) Page 2: "Decision trees for supervised machine learning problems have been proposed more than 40 years ago (Ritschard, 2010; Quinlan, 1986),"

   The statement that decision trees were proposed more than 40 years ago does not fit with the fact that the oldest reference the authors cite here is from 1986. How about citing the 1984 book "Classification and regressions trees"?

4) Page 2: "During inference, the observation vector is routed to a leaf node, and the contents of the leaf node are assigned as predictions."

   "the contents of the leaf node are assigned as predictions" is a bit vague. Oftentimes the end nodes are not pure. Usually one performs majority voting or (in the case of regression) takes the mean. Please improve this.

5) Page 3: "For further details on the key concepts of decision tree learning and tree-based machine learning, we refer to (Sheppard, 2017; Safavian and Landgrebe, 1991; Myles et al., 2004)."

   Please use "\citet" instead of "\citep" here.

6) Page 4: "With statistical significance, we demonstrate the emergence of the conditioning bias in both binary decision trees and random forests when attributes taking values from small, equidistant domains are present."

   It should be explained shortly what is meant by an "equidistant domain" first before using this (undefined?) term in the rest of the paper.

7) Page 3: "The proposed techniques and the completely reproducible implementation of the study are shared with the community in the GitHub repository https://github.com/gykovacs/conditioning bias."

   The programming language used should be stated.

8) Page 4: "Normal, boldface and calligraphic typesetting denote scalars ($d \in Z$), vectors ($x \in Zd$) and sets ($S \subset Z$), respectively."

Make clear that the expressions in brackets are examples.

9) Page 5: "The parameters ($f*$, $t*$) are recorded and left and right child nodes are added by applying step S1 to the training records indexed by L and R, respectively."

   Why is step S1 (checking the purity of the nodes) necessary to generate the child nodes? The child nodes are generated no matter whether they will be the end nodes or not.

10) Page 5: "For further details on decision tree induction and refer to (Sheppard, 2017)."

    The "and" should be removed here. Also: \citep

11) Page 5: "2.2 Inference with decision trees"

    "Inference" is not the correct term here. Statistical inference means something else typically (e.g. infering about the truth in presence of the sampling variability in the data), which is why using this term here can be confusing. "Prediction" should be used instead of "Inference" (not only in this title, but throughout the paper).

12) Page 5: "Upon encountering a leaf node, the recorded statistics (such as the class distribution for classification) are returned as the prediction for x."

    It's no always the class distribution. For point predictions usually the most frequent class in the lead node is used.

13) Page 6: "We also mention that due to the regularizing effect of bootstrap aggregation, the decision trees used in random forests need less regularization by depth,"

    Actually, they do not only not need this. It is even known that the trees in random forests should be grown to full size and this is also default implementation for classification in most RF implementations (paper: "Hyperparameters and tuning strategies for random forest").

14) Page 6: "For the ease of discussion, we introduce the term conditioning bias to refer to the variation of predictions due to the choice of conditioning."

    But a variation is not a bias. Please clarify this.

15) The authors write several times that they proposed method*s* (plural), but actually they only propose a single method.

16) Page 6: "To eliminate the potentially detrimental effects of the conditioning bias, we propose integrating it out by making predictions with both conditioning operators and averaging the results, as detailed in subsection 3.2."

    A natural way of dealing with this would be to make a random decision on which of the two child nodes the observation is assigned to for observations that are exactly on the split point (i.e., the mean between the neighboring values). This should at least be mentioned.

17) The paper features typos. It should be read carefully to fix these.

18) Page 9: "The benefit of tree mirroring is that predicting −x with the mirrored tree ..."

    This phrasing is confusing because it sounds as if one would predict the value of -x, which is obviously not the case.

19) Page 10: "We consider a feature susceptible to the conditioning bias if there exists an observed domain value xf and a positive real number r ∈ R+ such that xf – r and xf +r are also observed."

    That's only a minor thing, but wouldn't it be enough to demand that xf − r *or* xf + r are also observed?

20) Page 12: "Nevertheless, if the proposed method performs as expected relative to individual operators during crossvalidation, it is reasonable to assume this behavior persists when using all data for training."

    Here, the authors say themselves that the cross-validation-based prediction error estimates obtained for the individual operators is comparable to that expected when using the whole dataset for training. However, this means that the conditioning operator *can* be selected using cross-validation-based tuning.

21) Page 12: "Conversely, growing trees to their maximum depth might result in exaggerated estimations of the bias; however, these models are highly overfitted, leading to poor prediction results."

No, random forests for classification should be grown to maximum depth. This does not lead to overfitting here because the overfitting effects are "averaged out" across the trees.

22) Page 13: "Wilcoxon's signed-rank test, considered a generalization of the t-test,"

    The Wilcoxon test is not a "generalization of the t-test", but an alternative for the t-test for situations with metric, but non-normally distributed variables.

23) Page 13: "Finally, while the Wilcoxon test is related to the locations of the medians rather than the relation of the means, it is observed that on many folds there is no difference in the performance of two variations of an estimator, leading to identical medians. Therefore, as an indicative absolute measure, the mean performance scores and their differences are reported alongside the results of the hypothesis testing."

    It is not clear at all to me, how the second sentence relates to the first sentence. The "Therefore," starting the second sentence however suggests that there would be a relation between the two sentences.

24) Page 14: "The only exception is classification with decision trees, where fitting to all data eliminates biasing nodes in most cases."

    Speaking of the "only exception" here is misleading because there are only two options here (classification vs. regression).

25) Page 14: "This can be explained by considering that decision tree classification is the most strongly regularized problem in model selection, leading to trees with only 10-15 nodes in most cases."

    Apart from the fact that writing "the most strongly regularized problem in model selection" is confusing because it sounds like you would be speaking generally here about all problems in model selection, I cannot follow this sentence. If the problem is strongly regularized with few nodes, wouldn't this speak *against* the bias being strong because the bias occurs primarily in small nodes which occur more frequently in trees with many nodes?

26) Page 15: "Given that one cannot know which conditioning leads to better results, the results suggest that there is no more risk of using the proposed methods than using one particular conditioning."

Well, if the particular conditioning is the better performing condition, then the proposed method does lead to worse results than the particular conditioning.

27) Page 16: ", with popular implementations like (Pedregosa et al., 2011; Ripley, 2018)"

Please use \citep here.

28) Page 16: "In Section 3, we proposed the elimination of the bias by integrating out the dependence on conditioning,"

One should be careful with using mathematical terms like "integrating out" in situations where they do not exactly apply (it's not like the authors would exactly integrate something here).